

Exact Exponent Cylinders, Precision Budgets, and Singular Coding in Accelerated Syracuse Dynamics

Version 6

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Abstract

This paper continues the published v5 methodology retrospective [2]. It reports exact arithmetic results in the subsequent Lean development. Every nonempty positive-exponent word is characterized by one exact residue congruence, including the final odd endpoint bit; repeated blocks inherit a modulus growing with their length. A fixed expansive Syracuse exponent word consumes a precisely quantified number of 2-adic precision bits each time it matches an integer orbit. Compatible changes of affine phantom label preserve that decreasing counter; unrestricted changes can reset it. An explicit natural-number lasso demonstrates the failure of a proposed rank built from three short-word counters. A paper-level cancellation family shows that a constant affine defect need not bound destination precision. A separate theorem constructs, at every finite precision, positive odd inputs approaching the 2-adic singular point whose accelerated outputs remain different. We give the correct domain for infinite positive-exponent coding and record necessary cycle and finite-barrier constraints. These results identify a concrete certificate-search problem, but they do not prove strict descent for every positive integer or exclude every nontrivial cycle. The paper distinguishes Lean declarations, paper-level deductions, and open research targets.

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1 Scope and notation

The v5 paper [2] is a published retrospective of the earlier program. This version concerns the later arithmetic development. Let

$$S(n) = \frac{3n+1}{2^{a(n)}}, \quad a(n) = \nu_2(3n+1)$$

for a positive odd integer n . Every iterate of S is again a positive odd integer. For a finite word $w = (a_1, \dots, a_\ell)$ of positive exponents, put $A = \sum_i a_i$, $P = 3^\ell$, and $Q = 2^A$. Its affine constant C_w is defined by $C_\emptyset = 0$ and

$$C_{w\|(a)} = 3C_w + 2^\sum w.$$

When an orbit starting at n has exactly the first ℓ exponents w , its endpoint $m = S^\ell(n)$ satisfies

$$Qm = Pn + C_w. \tag{1}$$

This identity is formalized as `evalSyracuseWord_mul_pow_sum_eq_affine_of_matches` in `CollatzBridge.lean`. A finite word is *expansive* when $P > Q$. This refers to its affine slope, not to a claim that every individual matched endpoint exceeds its source.

2 Exact finite exponent cylinders

For a nonempty word w with every $a_i > 0$, let $\mathcal{C}_w$ be the set of natural starting values whose first ℓ accelerated Syracuse exponents are exactly w . The extra final bit in the following theorem distinguishes an odd endpoint from an integral but even one.

Theorem 1 (Exact cylinder congruence). *For every $n \in \mathbb{N}$,*

$$n \in \mathcal{C}_w \iff (3^\ell n + C_w) \bmod 2^{A+1} = 2^A. \tag{2}$$

If $k \geq 1$ and w^k denotes k consecutive copies of w , then

$$n \in \mathcal{C}_{w^k} \iff (3^{k\ell} n + C_{w^k}) \bmod 2^{kA+1} = 2^{kA}. \tag{3}$$

Proof. For a matching word, Equation (1) gives $3^\ell n + C_w = 2^A m$ with m odd, hence the stated residue. Conversely, the residue supplies an odd natural endpoint m satisfying this affine identity. Reverse induction through the positive exponent word, using the parity of each nonempty suffix constant, recovers each intermediate odd quotient and proves that every prescribed valuation is exact. Apply the same statement to w^k , whose exponent sum is kA , for Equation (3). $\square$

The Lean declarations are `syracuseWordMatches_iff_exactResidue` and `repeatPhantomWord_matches_iff_exactResidue` in `ExactCylinders.lean`; both pass the full library build. The final +1 matters: for $w = [1, 2]$, $A = 3$ and $C_w = 5$, so $n = 3$ makes $9n + 5 = 32$ divisible by 8 but does not realize the word, whereas $n = 11$ gives $9n + 5 = 104 \equiv 8 \pmod{16}$ and does realize it.

Corollary 2 (One residue class and its density; paper-level). *For each such w , $\mathcal{C}_w$ is one odd residue class modulo 2^{A+1} . Its natural density among positive integers is $2^{-(A+1)}$, or 2^{-A} relative to positive odd integers. The positive integer points of this class are dense within its 2-adic clopen cylinder, whose Haar measure in $\mathbb{Z}_2$ is $2^{-(A+1)}$. For w^k , replace A by kA .*

Proof. The coefficient 3^ℓ is invertible modulo 2^{A+1} , so Equation (2) has exactly one residue class. Since C_w is odd and 2^A is even, that class is odd. Counting one class per period gives the natural densities. Every 2-adic subcylinder of this class contains positive representatives of the same integer congruence, giving density within the cylinder. $\square$

The congruence theorem is formalized; uniqueness, density, and the topological description in the corollary are deductions in this paper. Ross’s earlier Zenodo v3 states the corresponding one-class result [7]; our contribution here is its exact Lean integration with the precision and trace developments, not a claim of mathematical priority.

3 An exact precision budget for one word

Fix a nonempty expansive word w . Write $D = P - Q > 0$ and

$$H_w(n) = Dn + C_w > 0, \quad p_w(n) = \nu_2(H_w(n)).$$

The rational formal fixed point of the word is $-C_w/D$. Since D is odd, $p_w(n)$ is also the 2-adic precision of the difference between n and that point.

Theorem 3 (Matched-word precision tax). *If w matches the orbit at n and has endpoint m , then*

$$QH_w(m) = PH_w(n), \quad p_w(m) + A = p_w(n). \quad (4)$$

If k consecutive copies of w match, then

$$kA \leq p_w(n) \leq \lfloor \log_2 H_w(n) \rfloor. \quad (5)$$

In particular, one fixed expansive word with $A > 0$ cannot repeat indefinitely along a positive integer orbit.

Proof. Multiply $H_w(m) = Dm + C_w$ by Q and use Equation (1):

$$Q(Dm + C_w) = D(Pn + C_w) + QC_w = P(Dn + C_w),$$

because $D + Q = P$. The multiplier $P = 3^\ell$ is odd, whereas $Q = 2^A$. Taking 2-adic valuations gives Equation (4); iteration telescopes, and nonnegativity of the final valuation gives Equation (5). The final upper bound is the elementary valuation-versus-bit-length inequality for a positive natural number. $\square$

The Lean declarations are `syracuseWord_precisionTax_scaled_identity_of_matches`, `syracuseWordPhantomPrecision_after_eval_add_sum`, `syracuseWordPhantomPrecision_after_iterate_add_mul_sum`, and `no_infinite_matched_expansive_word` in `PrecisionTax.lean`. `PrecisionTemplates.lean` supplies the words $[1]$, $[1, 2]$, and $[2, 1]$, for which $H_w(n)$ is respectively $n + 1$, $n + 5$, and $n + 7$. The theorem is a bound on *consecutive copies of the same matched word*. It does not bound arbitrary switching among different words.

4 Switching labels and a concrete obstruction

One can attempt to follow a changing formal point by assigning each state an affine label (D, C) and precision $\nu_2(Dn + C)$. Suppose an exact Syracuse step has exponent e and sends n to m :

$$2^e m = 3n + 1.$$

If source and destination labels obey

$$3D' = qD, \quad D' + 2^e C' = qC \quad (6)$$

for an odd factor q , then

$$2^e(D'm + C') = q(Dn + C). \quad (7)$$

For nonzero numerators this implies exact precision loss by e . This is `switchingPhantomPrecision_after_step_add_exponent` in `SwitchingPrecision.lean`.

The compatibility equations matter. At any odd n , a freely chosen shift $C_k = (2^k - 1)n$ gives $\nu_2(n + C_k) = k$ for every k . Thus a “choose the closest phantom afresh” rule need not decrease. This is the Lean theorem `arbitrary_phantom_shift_can_set_precision`.

Proposition 4 (A short-trace lasso). *The exact matched segment $743 \rightarrow 1115 \rightarrow 1255$ has exponent blocks $[1]$ and $[1, 2]$. The three-counter vector*

$$(\nu_2(n + 1), \nu_2(n + 5), \nu_2(n + 7))$$

takes the values

$$(3, 2, 1) \longrightarrow (2, 5, 1) \longrightarrow (3, 2, 1).$$

The compatible hidden counter $\nu_2(11n + 19)$ loses four bits across the full $[1, 1, 2]$ segment.

The segment and hidden loss are formalized by `concrete_shortTrace_lasso` and `hiddenTrace_lasso_tax` in `ThreeTraceObstruction.lean`. The recurrence $n_{k+1} = 1024n_k + 1767$ with $n_0 = 743$ produces arbitrarily precise positive integer approximants to the formal point $-19/11$; it does not produce an infinite positive integer orbit with a repeating state. The repeated triple rules out a strictly decreasing rank determined *solely* by those three counters; it does not rule out an augmented state or a rank tracking the compatible hidden counter. Moreover, `AffineTraceGraph.lean` proves that two prescribed words with different nondegenerate rational fixed points cannot both be self-cycles of one scalar rational control state (`no_common_rational_fixed_point_of_distinct_words`). These facts constrain a certificate search based on a small fixed set of phantom labels.

5 The singular point and the coding domain

Erratum to the published v4. The definition preceding the non-transport lemma in the published v4 [1] called the accelerated Syracuse map a continuous map $\tilde{S} : \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ and then called its valuation-word map a topological conjugacy with a shift. The first assertion is false, so the conjugacy argument as written is invalid. Its advertised global type is also wrong: an even input can have first exponent 0, outside the positive-exponent alphabet, while at $x = -1/3$ the numerator vanishes and has no finite valuation exponent. We replace those claims with the finite arithmetic witness below and restrict infinite-word coding to the invariant odd domain on which every numerator is nonzero. The logical non-transport observation for positive integer orbits survives by the direct arithmetic argument following the theorem; it requires no global continuous extension or global topological conjugacy.

The total function `Syracuse2adic` in `Syracuse2Adic.lean` uses the odd unit factor of $3x + 1$ and sets the value to 0 at $x = -1/3 \in \mathbb{Z}_2$. This total map agrees with S on positive odd integers, but it is not continuous at $-1/3$.

Theorem 5 (Finite-precision separation). *For every $R \in \mathbb{N}$, there are positive odd integers n_1, n_7 such that*

$$2^R \mid (3n_1 + 1), \quad 2^R \mid (3n_7 + 1), \quad S(n_1) = 1, \quad S(n_7) = 7.$$

Proof. Starting from 1 and 9, repeatedly apply $L(n) = 4n + 1$. The identity $3L(n) + 1 = 4(3n + 1)$ raises the valuation by two without changing the accelerated output. Thus, with

$$n_1(k) = \frac{4^{k+1} - 1}{3}, \quad n_7(k) = \frac{28 \cdot 4^k - 1}{3},$$

both positive odd sequences have the fixed outputs 1 and 7, while their numerators are divisible by 2^{2k+2} . Choose k large enough for the given R . $\square$

The Lean theorem is `singular_outputs_separated_at_every_precision` in `SyracuseSingularity.lean`. It asserts the finite arithmetic separation, not a topological result. Since 3 is a 2-adic unit, both sequences converge in $\mathbb{Z}_2$ to $-1/3$. Their different constant outputs then give the paper-level discontinuity conclusion.

Infinite positive-exponent words require a restricted domain. Let $U = \{x \in \mathbb{Z}_2 : x \text{ odd, } 3x + 1 \neq 0\}$, and define $D = \{x \in U : S^k(x) \in U \text{ for every } k \geq 0\}$, with S on U given by the odd unit factor of $3x + 1$. For $a \geq 1$ and odd $y \in \mathbb{Z}_2$, the inverse branch

$$\phi_a(y) = \frac{2^a y - 1}{3}$$

is odd, has valuation exponent a , and maps to y . Compositions for a fixed prefix of $W = (a_1, a_2, \dots)$ agree modulo $2^{a_1 + \dots + a_m}$. Their limit $\xi_W \in D$ is the unique point with that exponent word. In particular, a positive odd integer equals the point reconstructed from its own infinite word. The coding proof in this paragraph is paper-level; the current Lean module formalizes the arithmetic singularity witness, not this general reconstruction theorem.

Here *not eventually periodic* means that no tail of a word repeats with a fixed positive period. This qualification is essential: the word of $n = 3$ is $(1, 4, 2, 2, \dots)$, which is not periodic from the start but is eventually periodic. For *all* infinite positive-exponent words, the statement “ W is not eventually periodic $\implies \xi_W$ is not a positive integer” is equivalent to every positive odd integer having a bounded accelerated orbit. Indeed, a bounded positive orbit is pre-periodic and its exponent word is eventually periodic. Conversely, suppose a tail repeats a nonempty block w indefinitely. Then $P \neq Q$: $P = 3^{|w|}$ is odd and $Q = 2^A$ is even. The expansive case $P > Q$ is excluded for a positive integer by the matched-word precision budget of Theorem 3. In the remaining case $P < Q$, the block endpoint recurrence $m = (Pn + C_w)/Q$ has slope strictly below one and bounded iterates; its finitely many within-block intermediate states are bounded as well. Thus the integer orbit is bounded. This is the no-unbounded-orbit half of Collatz; the coding equivalence does not prove it. A claim restricted to a particular class of non-eventually-periodic words has different logical strength.

6 Related work and priority

Tao’s almost-bounded-value theorem is an almost-all statement in *logarithmic* density [3]. Inselsmann proves a different almost-all descent estimate in *natural* density on a logarithmic time scale [4]. Neither is a pointwise theorem for every positive integer. They set a stronger statistical context for the local finite-precision identities here, while leaving the universal bridge open.

Sharpe’s Lean repository reports near-periodicity and mechanical-itinerary results, among other formalized Collatz statements [5]. Fernández and Ibáñez study extremal parity words through Christoffel structures [6]. Ross’s Zenodo v3 establishes that a finite positive-exponent word occupies one odd residue class modulo $2^{1+\sum a_i}$ [7]. These sources overlap with the word and cycle language used here. The contribution claimed in this note is the *specific*

combination and formalization in this repository of the exact cylinder criterion, a matched-word precision budget, switching compatibility, and explicit obstructions. We make no claim of absolute mathematical priority for the underlying coding or valuation observations.

Recent AI-assisted claims include Mazur’s positive-density convergence formalization hosted by ProofAtlas [8] and Shaik’s proposed polylogarithmic descent for almost all starts [9]. ProofAtlas records a successful theorem-specific build while independent publication review remains pending. Shaik reports a Palomar Registry v2 replay of four selected Lean declarations; that selected replay does not by itself review the whole manuscript. The exact scope and evidence of both claims merit separate audit. We use neither as a premise, and neither presented claim gives a conclusion for every positive integer.

7 Cycle and finite-barrier constraints

If a nonempty word w matches a positive integer n and returns to n , Equation (1) gives

$$(2^A - 3^\ell)n = C_w. \quad (8)$$

Consequently $2^A > 3^\ell$, and any divisor of $2^A - 3^\ell$ divides C_w . These necessary statements are the Lean theorems `syracuseWordCycleEquationNat`, `syracuseWordCycleContracting`, and `syracuseWordCycleModularSieve` in `CycleConstraints.lean`. They do not show that all nontrivial cycles are impossible.

For a matched contracting finite word, let $d = 2^A - 3^\ell > 0$. Rearranging Equation (1) gives the exact displacement identity

$$2^A(n - m) = dn - C_w.$$

Hence $m < n$ if and only if $C_w < dn$. If the least nonnegative representative $r = n \bmod 2^{A+1}$ satisfies $C_w < dr$, then $r \leq n$ gives a finite residue certificate of strict descent. These are the scope of `evalSyracuseWord_lt_iff_source_above_barrier` and `evalSyracuseWord_lt_of_exactResidue_above_barrier` in `FirstBarrier.lean`. Evaluating the threshold at the actual source is algebraically equivalent to the desired drop; it is not an independent global ranking principle.

The related module `A0InfiniteSubfamily.lean` proves descent on the infinite residue classes $n = 615 + 4096u$ and $n = 2663 + 8192u$. This is a genuine pointwise result on two families, with no assertion that their union covers all positive odd integers.

8 A testable continuation

The results suggest a finite-certificate project with three separately checkable obligations:

1. Use Theorem 1 to enumerate finite exact exponent cylinders and attach each source residue either to a strict-descent certificate or to a declared control state.
2. On every continuing edge, verify a compatible affine label transport of Equation (6), or explicitly account for any counter reset or cancellation. The lasso of Proposition 4 invalidates a rank that records only the three shortest labels.
3. Prove that every possible exit and every uncovered residue has an actual descent witness; a finite matrix bound or a count of exceptional residues is not that witness.

The immediate falsifiable milestone is a small, exact certificate checker that reports uncovered residue classes and incompatible label cycles, not only successful finite examples. A theorem

covering all positive odd integers remains open. The named hypotheses `UniformStrictDescentHypothesis`, `A0FirstBarrierExistsAll`, and `A0FirstBarrierThresholdAutomatic` have not been proved. The last two together imply an exit for the A0 family only; the global descent bridge still needs a cover of every positive odd source.

A proposed defect and cancellation ledger (paper-level). The compatible switching identity can be extended to measure a failure of compatibility. On an exact edge $2^e m = 3n + 1$, assign integer affine labels $F_i(n) = D_i n + C_i$ and $F_j(m) = D_j m + C_j$, and choose an odd integer q . Define the edge defect

$$E(n) = (3D_j - qD_i)n + (D_j + 2^e C_j - qC_i).$$

Direct substitution gives

$$2^e F_j(m) = qF_i(n) + E(n).$$

When the compatibility equations (6) hold, E is identically zero and the verified switching identity applies. Otherwise extend ν_2 to signed integers by $\nu_2(t) = \nu_2(|t|)$ for $t \neq 0$ and $\nu_2(0) = +\infty$, and let $a = \nu_2(F_i(n))$, $b = \nu_2(E(n))$ when both are nonzero. The exact valuation ledger has three cases: if $a < b$, then $\nu_2(F_j(m)) + e = a$; if $b < a$, then $\nu_2(F_j(m)) + e = b$. If $a = b < \infty$, write $qF_i(n) = 2^a u$ and $E(n) = 2^a v$ with u, v odd, and put $c = \nu_2(u + v) \geq 1$. Then $\nu_2(F_j(m)) = a + c - e$, with the convention for a zero sum above. The proposed scalar precision rank drops strictly on this edge only if $c < e$; for $e = 1$, this is impossible. Cancellation can make the destination valuation arbitrarily higher, so the equal case must not be treated as a bounded reset.

Indeed, take $e = 1$, $q = 3$, $F_i(n) = n + 1$, and $F_j(m) = m + 5$. Then $E(n) = 8$, a constant with no 2-adic root. For every even $k \geq 4$,

$$n_k = \frac{2^{k+1} - 11}{3}, \quad m_k = S(n_k) = 2^k - 5$$

are positive odd integers in the fixed exact [1] cylinder $n_k \equiv 3 \pmod{4}$. Direct calculation gives $\nu_2(F_i(n_k)) = \nu_2(E(n_k)) = 3$ but $\nu_2(F_j(m_k)) = k$. Thus absence of a defect root does *not* bound destination precision. This family is a paper-level arithmetic counterexample, not a Lean declaration.

The defect still gives a useful exact cylinder test. On $x \equiv r \pmod{2^K}$, write $E(x) = sx + b_0$ with $s \neq 0$ and $\alpha = \nu_2(s)$. If $\nu_2(E(r)) < K + \alpha$, then $\nu_2(E(x)) = \nu_2(E(r))$ throughout the cylinder: write $x = r + 2^K t$ and compare the valuations of $E(r)$ and $s2^K t$. Otherwise E has a unique zero in that 2-adic cylinder, approximated by positive integer representatives with unbounded defect valuation. For $s = 0$, the defect is constant. A checker must also split the critical $a = b$ cylinders, track zeros of the pulled-back destination numerator $3D_j n + D_j + 2^e C_j$, or supply a separate descent certificate. The proposed falsifiable target is a finite exact-cylinder checker that reports both types of valuation tower and candidate non-descending control cycles at each tested resolution. No finite closure, global cover, or termination theorem follows from this ledger alone. The repository script `lean/scripts/phantom_taxonomy/switching_defect_probe.py` checks the identity, all three valuation cases, and the constant-defect family at finite depth. It is a reproducible diagnostic, not a proof of the proposed checker or a Lean formalization.

9 Verification boundary

The cited declarations are in the `lean/CollatzShadowing` library. The full library build completed on 24 September 2026 with Lean 4.29.1 and the matching Mathlib revision. There

are no source-level `sorry`, `admit`, or user-declared `axiom` declarations in that library. Some finite proofs use `native_decide`, which places native compilation in their trusted base; a source scan alone does not establish a kernel-only axioms claim. A `#print axioms` audit of selected structural headline theorems reports only standard Lean axioms.

The pinned toolchain predates the official Lean 4.32.2 kernel soundness fix [10] and further Lean 4.34.0 soundness fixes and checks [11]. No defect is known to be exercised by this development. A fresh build with Lean 4.34.0 and a compatible Mathlib version, followed by another axioms audit, is a separate verification task; this manuscript does not present it as completed.

10 Conclusion

The exact-cylinder theorem, finite-word precision tax, and compatible switching identity provide exact arithmetic progress measures. The trace lasso, incompatible cycle labels, cancellation family, and singularity witness delimit where simple globalizations fail. The cycle and barrier theorems supply necessary and local tests. Their joint value is a sharper specification for a future certificate that must cover every positive odd orbit. Neither half of the Collatz conjecture is settled here.

References

- [1] P. Borgatta, *Phantom Orbit Shadowing for the Collatz Conjecture: Verified Core, Two Barriers, and a Repositioning* (v4), Zenodo, [10.5281/zenodo.2054446](https://zenodo.org/record/2054446), 2026.
- [2] P. Borgatta, *Cross-AI, Lean-Verified Mathematics: A Case Study on the Collatz Conjecture* (v5), Zenodo, DOI [10.5281/zenodo.20554750](https://doi.org/10.5281/zenodo.20554750), 2026.
- [3] T. Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum of Mathematics, Pi 10 (2022), e12; <https://arxiv.org/abs/1909.03562>.
- [4] M. Inselmann, *An approximation of the Collatz map and a lower bound for the average total stopping time*, arXiv:2402.03276v3, 2024; <https://arxiv.org/abs/2402.03276v3>.
- [5] M. Sharpe, *Collatz at the Critical Line*, Lean repository, <https://github.com/msharpe248/collatz>, accessed 24 September 2026.
- [6] Fernández and Ibáñez, *Christoffel words as extremal structures in Collatz dynamics*, arXiv:2607.24844, 2026; <https://arxiv.org/abs/2607.24844>.
- [7] M. Ross, *Spike Structures and 2-adic Transition Laws in the Accelerated Collatz Map*, Zenodo v3, 30 August 2026; [10.5281/zenodo.22181823](https://zenodo.org/record/22181823).
- [8] L. Mazur, *Explicit Positive-Density Collatz Convergence in Logarithmic Time*, manuscript v2.1, 6 September 2026, ProofAtlas release; [formalization page](#) (independent publication review pending).
- [9] I. A. Shaik, *Polylogarithmic Descent for Almost All Collatz Orbits in Natural Density*, Zenodo v3.2.4, [10.5281/zenodo.22130385](https://zenodo.org/record/22130385), 2026; selected-declaration replay described at [the author's evidence page](#).
- [10] Lean Language Reference, *Lean 4.32.2 release notes*, <https://lean-lang.org/doc/reference/latest/releases/v4.32.2/>, 2026.
- [11] Lean Language Reference, *Lean 4.34.0 release notes*, <https://lean-lang.org/doc/reference/latest/releases/v4.34.0/>, 2026.